Interval-2 cooldowns, joint 5-interval search and per-LR lookaheads: live results

Updated 2026-10-23 19:17 PDT (02:17 UTC). BS 2000, CIFAR-10 airbench, score = TTA test accuracy, one training seed (0). Jobs: 56 done, 0 running, 0 pending, 0 failed.

1. Conv interval 2: more cooldown shapes and lengths

Setup. The one cell of Section 3 where every cooldown failed: no transform, conv, interval 2 (steps 40-80), candidates 0.047-0.36, where the truth peaks at 0.13 and every one-interval cooldown picked 0.078. Each candidate is trained for the interval from the plan's state and scored after a cooldown of 20, 40, 60 or 80 steps (t = i / length, i = 0 .. length-1), with the other LRs on the plan unless stated. Old shapes as in Section 3; new, conv only: **power** c (1-t)^p (p = 1 is linear to 0), **exponential** c q^t (from c to q c), **power to end q** c (q + (1-q)(1-t)^p). p is searched on the 0.6^k grid from 1, q and m from 0.22, each unbounded, moving while a k±1 or k±2 neighbour is better. The truth is recomputed in every job. One seed (0).

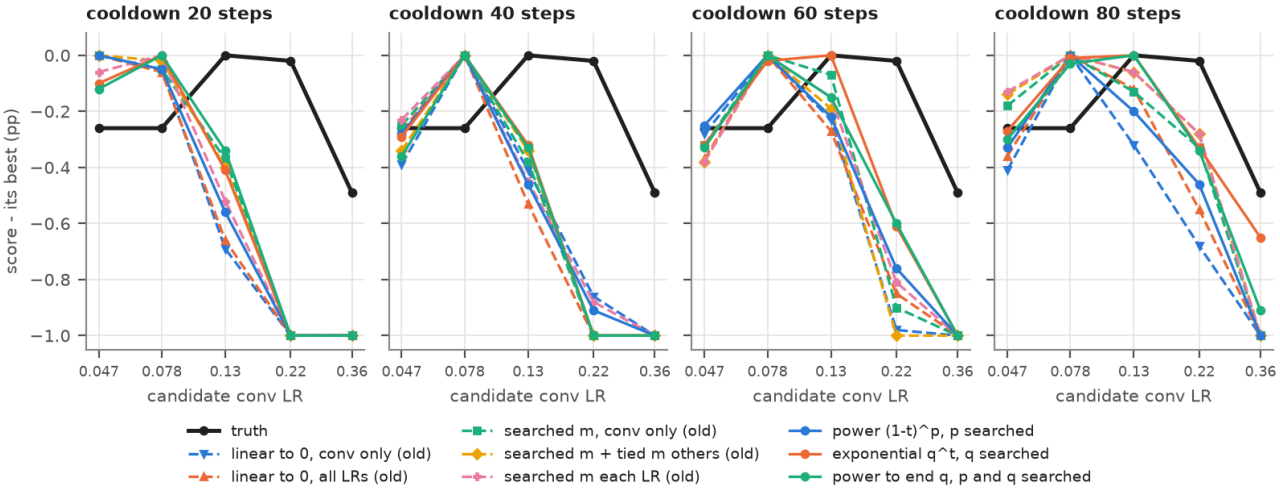


Figure 1. Score per candidate minus its own best (clipped at -1 pp), per cooldown length; a cooldown decides like the truth when it peaks where the truth does. Dashed: the Section 3 shapes; solid: the new ones.

Table 1. Candidate each cooldown picks and, in brackets, the truth it loses against the truth's best (pp); bold = the truth's pick. Truth per candidate (recomputed per job): 20 steps: 0.047 93.84, 0.078 93.84, 0.13 94.10, 0.22 94.08, 0.36 93.61; 40 steps: 0.047 93.84, 0.078 93.84, 0.13 94.10, 0.22 94.08, 0.36 93.61; 60 steps: 0.047 93.84, 0.078 93.84, 0.13 94.10, 0.22 94.08, 0.36 93.61; 80 steps: 0.047 93.84, 0.078 93.84, 0.13 94.10, 0.22 94.08, 0.36 93.61.

Cooldown	20 steps	40 steps	60 steps	80 steps
truth	**0.13 (0.00)**	**0.13 (0.00)**	**0.13 (0.00)**	**0.13 (0.00)**
linear to 0, conv only (old)	0.047 (0.26)	0.078 (0.26)	0.078 (0.26)	0.078 (0.26)
linear to 0, all LRs (old)	0.047 (0.26)	0.078 (0.26)	0.078 (0.26)	0.078 (0.26)
searched m, conv only (old)	0.047 (0.26)	0.078 (0.26)	0.078 (0.26)	0.078 (0.26)
searched m + tied m others (old)	0.047 (0.26)	0.078 (0.26)	0.078 (0.26)	0.078 (0.26)
searched m each LR (old)	0.078 (0.26)	0.078 (0.26)	0.078 (0.26)	0.078 (0.26)
power (1-t)^p, p searched	0.047 (0.26)	0.078 (0.26)	0.078 (0.26)	0.078 (0.26)
exponential q^t, q searched	0.078 (0.26)	0.078 (0.26)	**0.13 (0.00)**	**0.13 (0.00)**
power to end q, p and q searched	0.078 (0.26)	0.078 (0.26)	0.078 (0.26)	**0.13 (0.00)**

Table 1p. Searched cooldown parameters per candidate.

Shape, length	c = 0.047	c = 0.078	c = 0.13	c = 0.22	c = 0.36
searched m, conv only (old), 20	conv=0.22	conv=0.13	conv=0.22	conv=0.22	conv=0.13
searched m, conv only (old), 40	conv=0.36	conv=0.22	conv=0.22	conv=0.13	conv=0.13
searched m, conv only (old), 60	conv=0.6	conv=0.22	conv=0.13	conv=0.13	conv=0.13
searched m, conv only (old), 80	conv=0.36	conv=0.13	conv=0.13	conv=0.13	conv=0.13
power (1-t)^p, p searched, 20	p=1.7	p=1.7	p=2.8	p=1.7	p=1.7
power (1-t)^p, p searched, 40	p=0.36	p=0.6	p=1	p=1	p=1.7
power (1-t)^p, p searched, 60	p=0.6	p=0.6	p=1.7	p=1.7	p=1.7
power (1-t)^p, p searched, 80	p=0.6	p=1	p=1.7	p=1.7	p=1.7
exponential q^t, q searched, 20	q=0.078	q=0.078	q=0.13	q=0.078	q=0.047
exponential q^t, q searched, 40	q=0.36	q=0.13	q=0.078	q=0.028	q=0.047
exponential q^t, q searched, 60	q=0.13	q=0.13	q=0.078	q=0.047	q=0.028
exponential q^t, q searched, 80	q=0.078	q=0.047	q=0.047	q=0.028	q=0.047
power to end q, p and q searched, 20	p=2.8, q=0.22	p=2.8, q=0.13	p=2.8, q=0.13	p=4.6, q=0.13	p=2.8, q=0.078
power to end q, p and q searched, 40	p=1, q=0.22	p=1.7, q=0.22	p=4.6, q=0.13	p=1.7, q=0.047	p=2.8, q=0.078
power to end q, p and q searched, 60	p=1, q=0.047	p=1, q=0.13	p=2.8, q=0.13	p=1.7, q=0.047	p=4.6, q=0.078
power to end q, p and q searched, 80	p=1.7, q=0.078	p=1.7, q=0.078	p=1.7, q=0.047	p=2.8, q=0.078	p=2.8, q=0.047

2. Capacity of the 5-constant-interval conv schedule: joint neighbourhood search

Setup. Conv LR only (the other groups on the baseline); conv's specified schedule is 5 constant intervals on the x0.6 grid, and the transform multiplies conv's applied LR. One search seed (0). Each iteration trains all 3^5 - 1 = 242 joint neighbours (every interval -1, 0 or +1 grid step) as full runs, moves to the best if it beats the current schedule, and repeats until none does. Start: the oracle (baseline / multiplier) reduced to its 5 interval means, snapped to the grid. The result is the winner's search score (no replay); baseline and do-nothing are single runs on the same seed.

Table 2. Joint search per transform. Baseline (no transform, seed 0): 93.92. Iterations = moves made so far; best evaluated = the best full run so far (the schedule the search moves to next); a running search's best is final only once it converges.

Transform	do nothing	start (score)	best evaluated (score)	Δ vs baseline	iterations	runs	status
no transform	93.92	0.22, 0.13, 0.13, 0.078, 0.028 (93.53)	0.078, 0.13, 0.078, 0.047, 0.01 (94.09)	+0.17	3	682	converged
ramp x0.25->x4	93.31	0.6, 0.22, 0.13, 0.047, 0.01 (93.13)	0.36, 0.36, 0.078, 0.047, 0.006 (93.88)	-0.04	1	438	converged
reverse ramp x4->x0.25	93.47	0.078, 0.078, 0.13, 0.13, 0.078 (93.61)	0.078, 0.078, 0.047, 0.047, 0.017 (94.17)	+0.25	3	711	converged
wave x2^sin	93.68	0.13, 0.078, 0.13, 0.13, 0.047 (93.19)	0.078, 0.078, 0.13, 0.13, 0.017 (93.93)	+0.01	2	531	converged
inverted wave	93.24	0.36, 0.36, 0.13, 0.047, 0.017 (93.73)	0.36, 0.36, 0.078, 0.028, 0.01 (94.03)	+0.11	1	414	converged
steep ramp x1/8->x8	92.47	1, 0.36, 0.13, 0.028, 0.006 (93.07)	1, 0.22, 0.078, 0.01, 0.0013 (93.84)	-0.08	3	744	converged

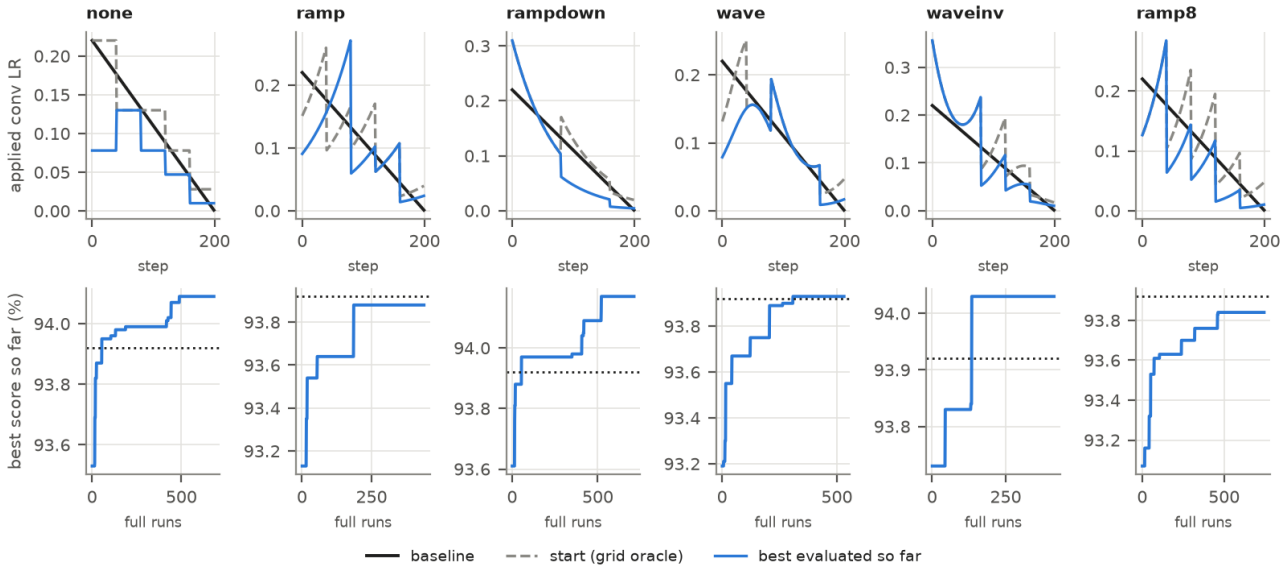


Figure 2. Top: applied conv LR of the start and the best evaluated schedule against the baseline. Bottom: best score among the full runs so far (dotted: baseline, seed 0).

Interval-by-interval refinement (ICS pass 2) on the same plan. Same schedule class, transforms and seed; the plan is the joint search's start (no shift). For each interval in order, the first included, from the state after the intervals already committed: score the plan's value, then one grid step smaller and, only if that is not better, one grid step larger; a candidate's score is the interval at the candidate plus the rest of the run on the plan (a full run's TTA). The best is committed. No two-continuation average for the last interval; the result is the last committed score (no replay). The joint search's first iteration trained every schedule within one grid step of the same plan, so its best there ("best in the box") is what a perfect one-step search could reach (research/plan_refine_search.py).

Table 2b. Interval refinement against the joint search (TTA %, seed 0). Rank in box = the refinement's rank among the 243 schedules within one grid step of the plan (all trained by the joint search). Runs = training steps / 200; the refinement re-scores the committed value at each interval.

Transform	do nothing	plan	refinement (score)	rank in box	best in ±1 box (score)	joint search (score)	runs: refine / joint
no transform	93.92	93.53	0.13, 0.22, 0.078, 0.047, 0.028 (93.99)	1 / 243	0.13, 0.22, 0.078, 0.047, 0.028 (93.99)	0.078, 0.13, 0.078, 0.047, 0.01 (94.09)	7.0 / 682
ramp x0.25->x4	93.31	93.13	0.36, 0.13, 0.13, 0.028, 0.006 (93.52)	15 / 243	0.36, 0.36, 0.078, 0.047, 0.006 (93.88)	0.36, 0.36, 0.078, 0.047, 0.006 (93.88)	6.6 / 438
reverse ramp x4->x0.25	93.47	93.61	0.047, 0.047, 0.13, 0.13, 0.078 (93.95)	3 / 243	0.13, 0.078, 0.078, 0.078, 0.047 (93.97)	0.078, 0.078, 0.047, 0.047, 0.017 (94.17)	7.2 / 711
wave x2^sin	93.68	93.19	0.078, 0.078, 0.13, 0.078, 0.028 (93.89)	1 / 243	0.078, 0.078, 0.13, 0.078, 0.028 (93.89)	0.078, 0.078, 0.13, 0.13, 0.017 (93.93)	7.4 / 531
inverted wave	93.24	93.73	0.36, 0.36, 0.13, 0.028, 0.01 (93.81)	6 / 243	0.36, 0.36, 0.078, 0.028, 0.01 (94.03)	0.36, 0.36, 0.078, 0.028, 0.01 (94.03)	8.4 / 414
steep ramp x1/8->x8	92.47	93.07	0.6, 0.22, 0.13, 0.028, 0.0036 (93.52)	8 / 243	0.6, 0.22, 0.078, 0.028, 0.0036 (93.70)	1, 0.22, 0.078, 0.01, 0.0013 (93.84)	7.0 / 744

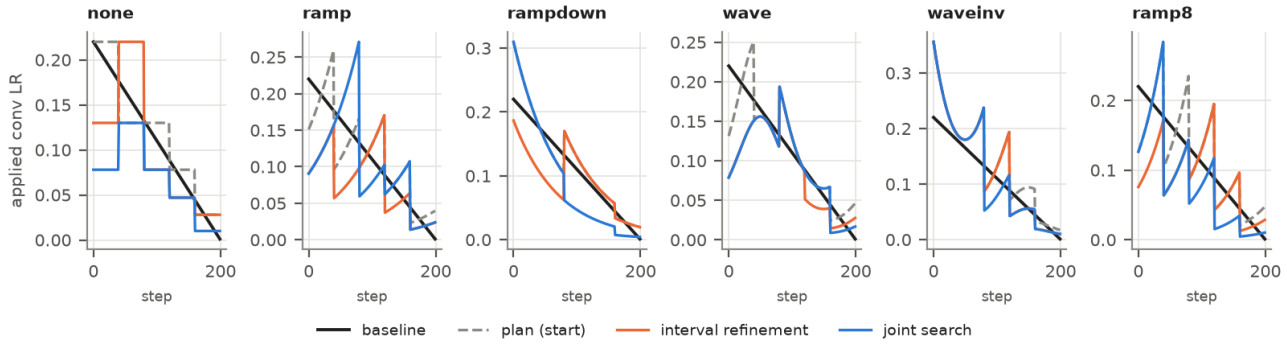


Figure 2b. Applied conv LR: baseline, plan, the interval refinement's schedule and the joint search's winner.

3. Which one-interval lookahead ranks interval candidates like the truth?

Setup. The plan is the baseline run itself, split into 5 intervals of 40 steps. Under a transform, the multiplier applies to all four LR groups, and the plan is the oracle (each group's baseline divided by the multiplier), so the plan's applied LRs are the baseline's. For a main LR and interval k, the plan is followed up to k; there the main LR is replaced by a constant candidate c on the x0.6 grid (the grid value nearest the plan's mean over the interval, and 1 and 2 grid steps either side; whitening bias, whose plan is 0 from step 75, uses its step-0 plan value), trained for one interval from the same state, and scored by one more interval of: **linear to 0, main LR only** (c to 0, the others on the plan); **linear to 0, all LRs**; **searched m, main LR only** (constant m x c); **searched m main + tied m others** (the other three at m' x their plan mean over interval k, one m' for all three); **searched m for each LR**. The **truth** is the final score when the run continues on the plan. Each m is searched on the 0.6^k grid without bounds, from 0.22, one multiplier at a time, moving while a neighbour at k±1 or k±2 is better; the result beats all four neighbours. A lookahead decides like the truth when its curve peaks at the same candidate. One seed: a single-run score has an SD of about 0.12 pp, so differences below ~0.15 pp between candidates are within noise.

3.1 no transform

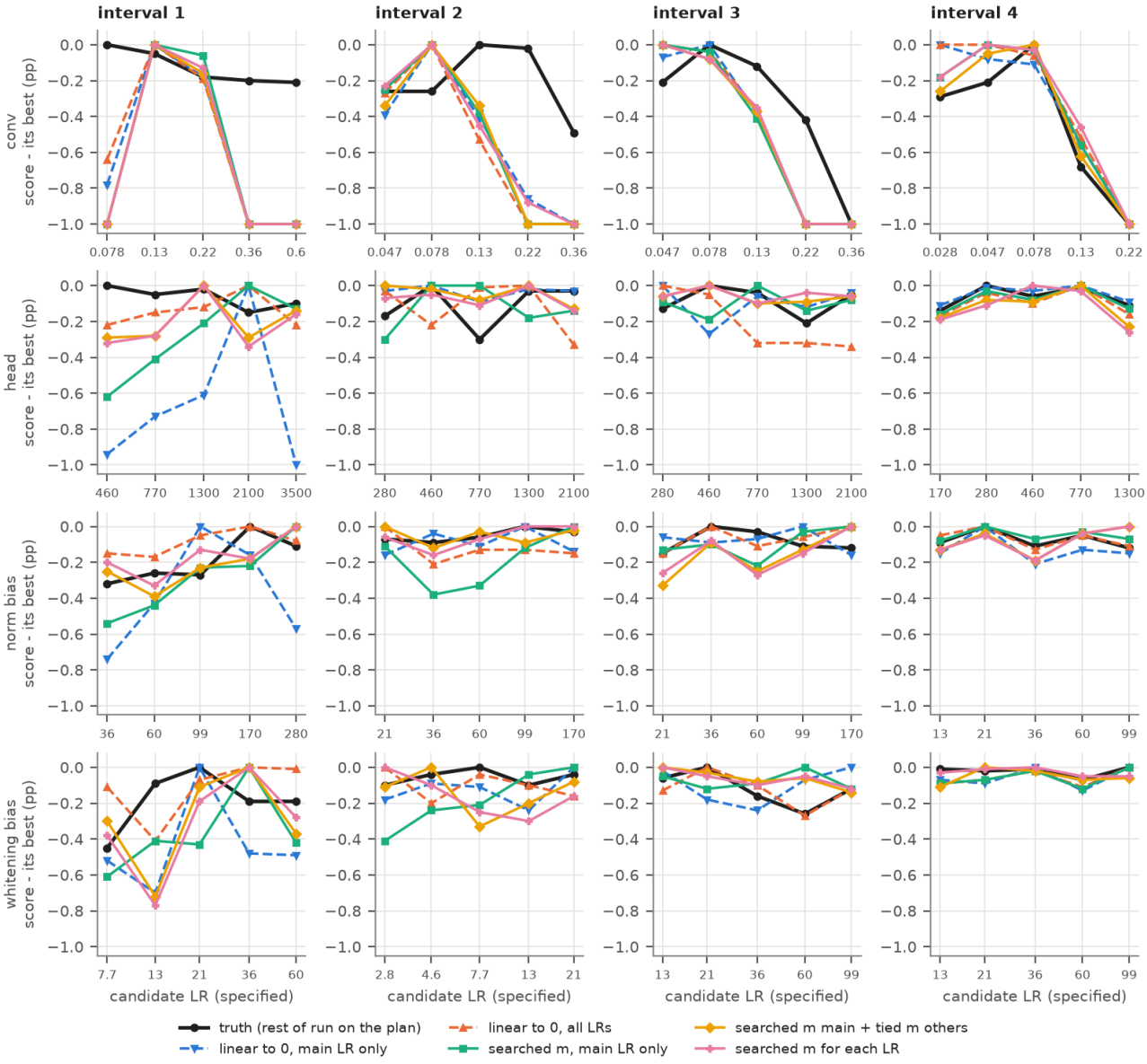


Figure 3.1. no transform: each score per candidate minus its own best (clipped at -1 pp); rows are the main LR, columns the interval.

Table 3.1. Candidate each lookahead picks and, in brackets, the truth it loses against the truth's best (pp); bold = the truth's pick.

LR, interval	truth best	linear to 0, main LR only	linear to 0, all LRs	searched m, main LR only	searched m main + tied m others	searched m for each LR
conv, 1	0.078	0.13 (0.05)	0.13 (0.05)	0.13 (0.05)	0.13 (0.05)	0.13 (0.05)
conv, 2	0.13	0.078 (0.26)	0.078 (0.26)	0.078 (0.26)	0.078 (0.26)	0.078 (0.26)
conv, 3	0.078	**0.078 (0.00)**	0.047 (0.21)	0.047 (0.21)	0.047 (0.21)	0.047 (0.21)
conv, 4	0.078	0.028 (0.29)	0.028 (0.29)	0.047 (0.21)	**0.078 (0.00)**	0.047 (0.21)
head, 1	460	2100 (0.15)	2100 (0.15)	2100 (0.15)	1300 (0.02)	1300 (0.02)
head, 2	460	**460 (0.00)**	1300 (0.03)	**460 (0.00)**	280 (0.17)	1300 (0.03)
head, 3	460	280 (0.13)	280 (0.13)	770 (0.04)	**460 (0.00)**	**460 (0.00)**
head, 4	280	770 (0.01)	770 (0.01)	770 (0.01)	770 (0.01)	460 (0.06)
norm bias, 1	170	99 (0.27)	**170 (0.00)**	280 (0.11)	280 (0.11)	280 (0.11)
norm bias, 2	99	**99 (0.00)**	21 (0.07)	170 (0.03)	21 (0.07)	**99 (0.00)**
norm bias, 3	36	99 (0.11)	**36 (0.00)**	170 (0.12)	170 (0.12)	170 (0.12)
norm bias, 4	21	**21 (0.00)**	**21 (0.00)**	**21 (0.00)**	99 (0.12)	99 (0.12)
whitening bias, 1	21	**21 (0.00)**	36 (0.19)	36 (0.19)	36 (0.19)	36 (0.19)
whitening bias, 2	7.7	21 (0.04)	2.8 (0.10)	21 (0.04)	4.6 (0.04)	2.8 (0.10)
whitening bias, 3	21	99 (0.12)	**21 (0.00)**	60 (0.26)	13 (0.06)	13 (0.06)
whitening bias, 4	99	36 (0.01)	**99 (0.00)**	**99 (0.00)**	21 (0.02)	36 (0.01)

Table 3.1m. Searched cooldown multipliers over all candidates of intervals 1-4: how often 0.22 is chosen, and the most common other values.

Main LR	searched m, main only	main + tied: main m	main + tied: others' m	each: main m
conv	0.22 in 9/20; else 0.13: 6, 0.36: 3, 0.078: 1	0.22 in 8/20; else 0.13: 6, 0.36: 5, 0.078: 1	0.22 in 3/20; else 0.6: 5, 1.7: 3, 0.078: 3	0.22 in 7/20; else 0.13: 7, 0.36: 5, 0.078: 1
head	0.22 in 2/20; else 0.36: 5, 0.6: 4, 0.078: 4, 0.13: 2	0.22 in 3/20; else 0.13: 4, 0.36: 4, 0.078: 3, 0.6: 3	0.22 in 16/20; else 0.13: 4	0.22 in 4/20; else 0.36: 5, 0.13: 4, 0.6: 4
norm bias	0.22 in 1/20; else 0.36: 6, 1: 3, 0.078: 3, 0.13: 2	0.22 in 5/20; else 0.6: 4, 0.13: 3, 0.36: 3	0.22 in 19/20; else 0.13: 1	0.22 in 5/20; else 0.6: 6, 0.13: 3, 0.36: 3
whitening bias	0.22 in 5/20; else 0.6: 4, 0.13: 3, 0.078: 2	0.22 in 4/20; else 0.6: 6, 0.36: 4, 1.7: 2	0.22 in 17/20; else 0.13: 3	0.22 in 6/20; else 0.36: 6, 0.6: 5, 0.078: 1

3.2 ramp x0.25->x4

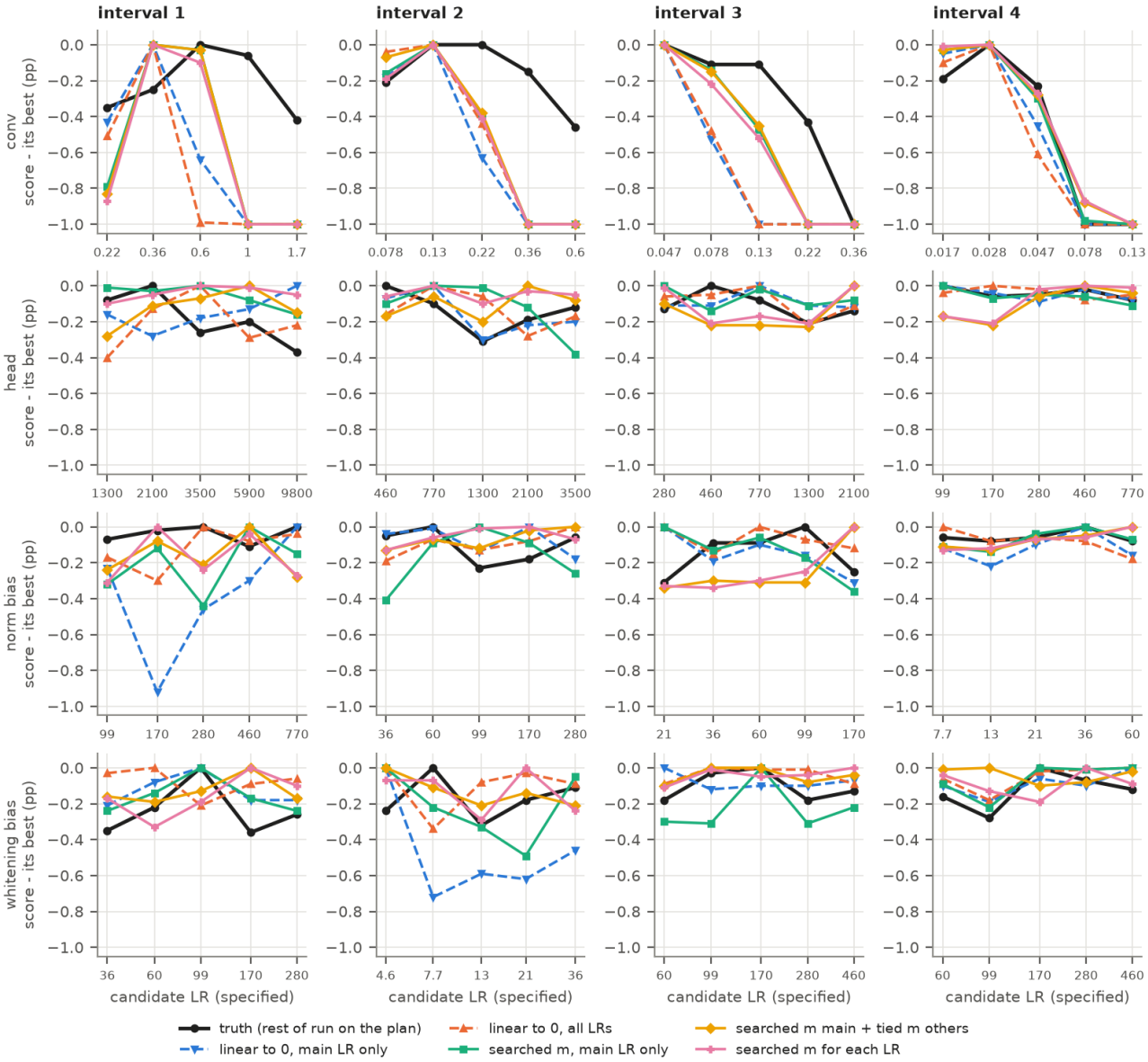


Figure 3.2. ramp x0.25->x4: each score per candidate minus its own best (clipped at -1 pp); rows are the main LR, columns the interval.

Table 3.2. Candidate each lookahead picks and, in brackets, the truth it loses against the truth's best (pp); bold = the truth's pick.

LR, interval	truth best	linear to 0, main LR only	linear to 0, all LRs	searched m, main LR only	searched m main + tied m others	searched m for each LR
conv, 1	0.6	0.36 (0.25)	0.36 (0.25)	0.36 (0.25)	0.36 (0.25)	0.36 (0.25)
conv, 2	0.13	**0.13 (0.00)**	**0.13 (0.00)**	**0.13 (0.00)**	**0.13 (0.00)**	**0.13 (0.00)**
conv, 3	0.047	**0.047 (0.00)**	**0.047 (0.00)**	**0.047 (0.00)**	**0.047 (0.00)**	**0.047 (0.00)**
conv, 4	0.028	**0.028 (0.00)**	**0.028 (0.00)**	**0.028 (0.00)**	**0.028 (0.00)**	**0.028 (0.00)**
head, 1	2100	9800 (0.37)	3500 (0.26)	3500 (0.26)	5900 (0.20)	3500 (0.26)
head, 2	460	770 (0.10)	770 (0.10)	770 (0.10)	2100 (0.19)	770 (0.10)
head, 3	460	770 (0.08)	770 (0.08)	280 (0.13)	2100 (0.14)	2100 (0.14)
head, 4	99	**99 (0.00)**	170 (0.06)	**99 (0.00)**	460 (0.02)	460 (0.02)
norm bias, 1	280	770 (0.00)	**280 (0.00)**	460 (0.11)	460 (0.11)	170 (0.02)
norm bias, 2	60	170 (0.18)	280 (0.06)	99 (0.23)	280 (0.06)	170 (0.18)
norm bias, 3	99	21 (0.31)	21 (0.31)	21 (0.31)	170 (0.25)	170 (0.25)
norm bias, 4	36	**36 (0.00)**	7.7 (0.06)	**36 (0.00)**	60 (0.08)	60 (0.08)
whitening bias, 1	99	**99 (0.00)**	60 (0.22)	**99 (0.00)**	170 (0.36)	170 (0.36)
whitening bias, 2	7.7	4.6 (0.24)	4.6 (0.24)	4.6 (0.24)	4.6 (0.24)	21 (0.18)
whitening bias, 3	170	60 (0.18)	99 (0.03)	**170 (0.00)**	99 (0.03)	460 (0.13)
whitening bias, 4	170	460 (0.12)	460 (0.12)	**170 (0.00)**	99 (0.28)	280 (0.07)

Table 3.2m. Searched cooldown multipliers over all candidates of intervals 1-4: how often 0.22 is chosen, and the most common other values.

Main LR	searched m, main only	main + tied: main m	main + tied: others' m	each: main m
conv	0.22 in 2/20; else 0.13: 9, 0.078: 8, 0.047: 1	0.22 in 3/20; else 0.078: 10, 0.13: 6, 0.047: 1	0.22 in 6/20; else 0.6: 5, 0.36: 5, 1: 1	0.22 in 3/20; else 0.078: 11, 0.13: 6
head	0.22 in 6/20; else 0.078: 5, 1: 3, 0.6: 2	0.22 in 0/20; else 0.36: 7, 1.7: 3, 0.13: 3, 0.078: 3	0.22 in 0/20; else 0.078: 17, 0.13: 2, 0.047: 1	0.22 in 3/20; else 0.078: 4, 0.36: 3, 0.6: 3
norm bias	0.22 in 6/20; else 0.36: 4, 0.6: 3, 0.078: 3	0.22 in 7/20; else 0.078: 4, 0.13: 3, 0.6: 2	0.22 in 0/20; else 0.078: 17, 0.13: 2, 0.047: 1	0.22 in 4/20; else 0.36: 4, 0.13: 4, 0.078: 3
whitening bias	0.22 in 6/20; else 0.13: 5, 0.078: 3, 1: 2	0.22 in 1/20; else 0.6: 7, 0.13: 4, 0.047: 2, 0.36: 2	0.22 in 0/20; else 0.078: 17, 0.13: 2, 0.047: 1	0.22 in 4/20; else 0.078: 4, 0.6: 4, 0.36: 3

3.3 reverse ramp x4->x0.25

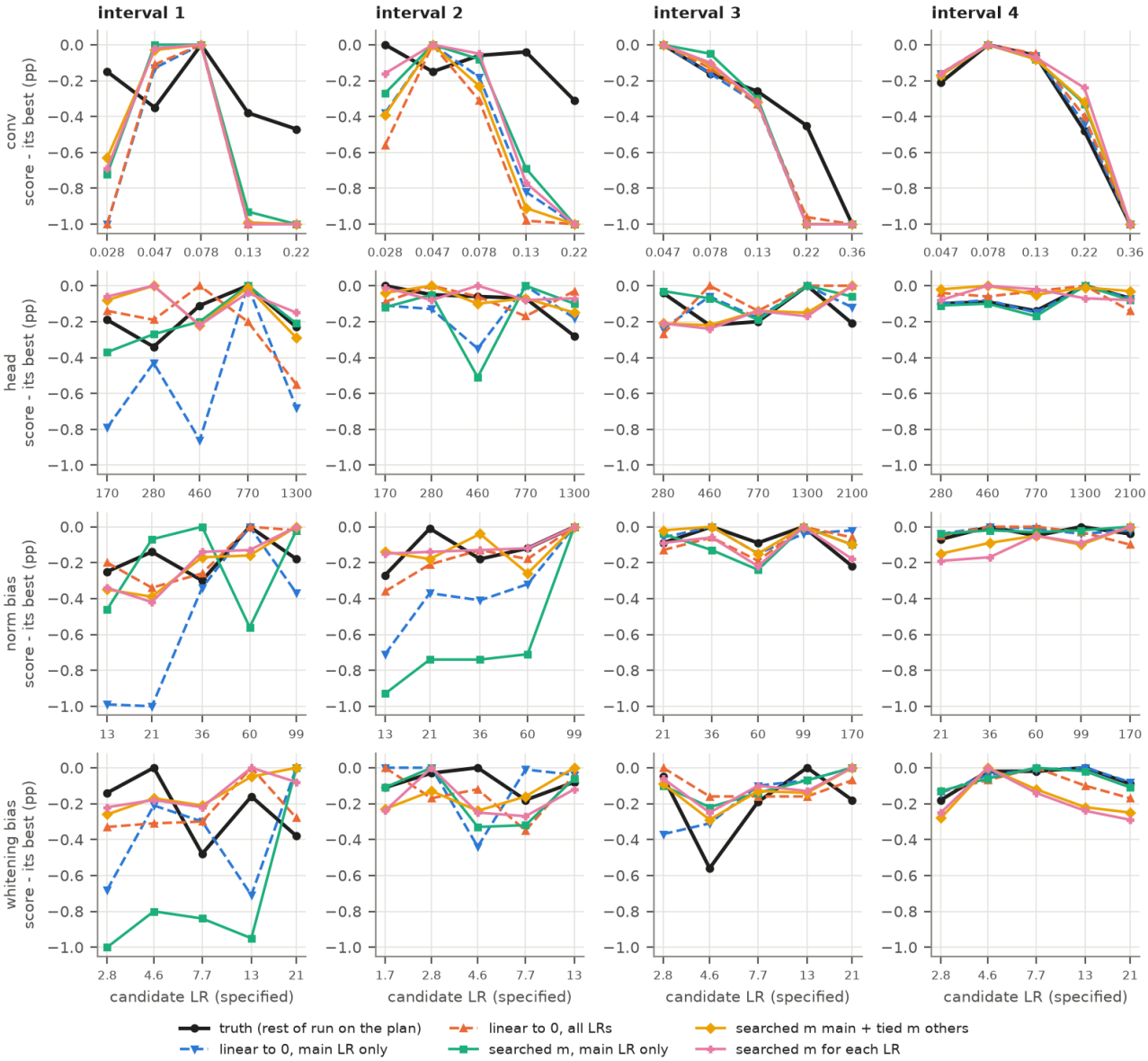


Figure 3.3. reverse ramp x4->x0.25: each score per candidate minus its own best (clipped at -1 pp); rows are the main LR, columns the interval.

Table 3.3. Candidate each lookahead picks and, in brackets, the truth it loses against the truth's best (pp); bold = the truth's pick.

LR, interval	truth best	linear to 0, main LR only	linear to 0, all LRs	searched m, main LR only	searched m main + tied m others	searched m for each LR
conv, 1	0.078	**0.078 (0.00)**	**0.078 (0.00)**	0.047 (0.35)	**0.078 (0.00)**	**0.078 (0.00)**
conv, 2	0.028	0.047 (0.15)	0.047 (0.15)	0.047 (0.15)	0.047 (0.15)	0.047 (0.15)
conv, 3	0.047	**0.047 (0.00)**	**0.047 (0.00)**	**0.047 (0.00)**	**0.047 (0.00)**	**0.047 (0.00)**
conv, 4	0.078	**0.078 (0.00)**	**0.078 (0.00)**	**0.078 (0.00)**	**0.078 (0.00)**	**0.078 (0.00)**
head, 1	770	**770 (0.00)**	460 (0.11)	**770 (0.00)**	280 (0.34)	280 (0.34)
head, 2	170	770 (0.07)	280 (0.05)	770 (0.07)	280 (0.05)	460 (0.06)
head, 3	1300	**1300 (0.00)**	460 (0.22)	**1300 (0.00)**	2100 (0.21)	2100 (0.21)
head, 4	1300	**1300 (0.00)**	**1300 (0.00)**	**1300 (0.00)**	460 (0.09)	460 (0.09)
norm bias, 1	60	**60 (0.00)**	**60 (0.00)**	36 (0.30)	99 (0.18)	99 (0.18)
norm bias, 2	99	**99 (0.00)**	**99 (0.00)**	**99 (0.00)**	**99 (0.00)**	**99 (0.00)**
norm bias, 3	36	**36 (0.00)**	99 (0.00)	99 (0.00)	**36 (0.00)**	99 (0.00)
norm bias, 4	36	**36 (0.00)**	**36 (0.00)**	170 (0.04)	170 (0.04)	170 (0.04)
whitening bias, 1	4.6	21 (0.38)	13 (0.16)	21 (0.38)	21 (0.38)	13 (0.16)
whitening bias, 2	4.6	1.7 (0.11)	1.7 (0.11)	2.8 (0.03)	13 (0.08)	2.8 (0.03)
whitening bias, 3	13	21 (0.18)	2.8 (0.05)	21 (0.18)	21 (0.18)	21 (0.18)
whitening bias, 4	13	**13 (0.00)**	7.7 (0.02)	7.7 (0.02)	4.6 (0.02)	4.6 (0.02)

Table 3.3m. Searched cooldown multipliers over all candidates of intervals 1-4: how often 0.22 is chosen, and the most common other values.

Main LR	searched m, main only	main + tied: main m	main + tied: others' m	each: main m
conv	0.22 in 5/20; else 0.36: 9, 0.6: 4, 1: 2	0.22 in 6/20; else 0.36: 8, 0.6: 5, 1: 1	0.22 in 6/20; else 0.13: 4, 0.36: 3, 4.6: 2	0.22 in 5/20; else 0.36: 8, 0.6: 6, 1: 1
head	0.22 in 3/20; else 0.13: 5, 0.36: 4, 0.6: 3	0.22 in 3/20; else 0.36: 4, 2.8: 3, 0.6: 2	0.22 in 2/20; else 0.36: 17, 0.6: 1	0.22 in 2/20; else 0.6: 5, 0.078: 4, 0.36: 4, 1: 3
norm bias	0.22 in 7/20; else 0.36: 6, 0.078: 3, 7.7: 1	0.22 in 3/20; else 0.13: 4, 7.7: 3, 1: 3, 1.7: 3	0.22 in 2/20; else 0.36: 16, 0.6: 2	0.22 in 2/20; else 0.078: 4, 0.6: 4, 0.36: 3
whitening bias	0.22 in 1/20; else 0.6: 6, 0.13: 5, 0.078: 3, 0.36: 3	0.22 in 4/20; else 0.6: 5, 0.078: 3, 0.13: 2	0.22 in 0/20; else 0.36: 19, 0.6: 1	0.22 in 4/20; else 0.6: 7, 0.36: 3, 0.13: 3

3.4 wave x2^sin

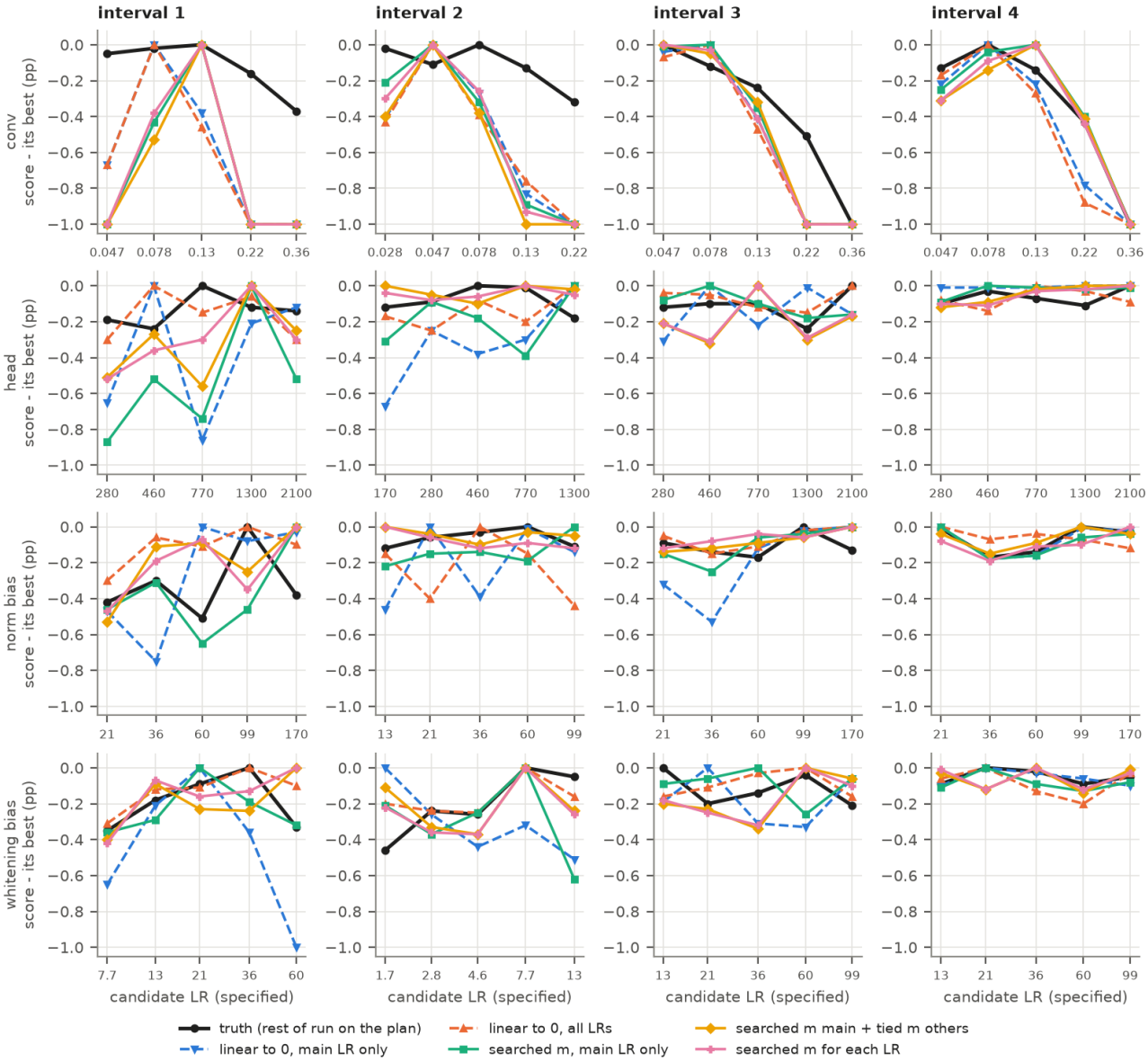


Figure 3.4. wave x2^sin: each score per candidate minus its own best (clipped at -1 pp); rows are the main LR, columns the interval.

Table 3.4. Candidate each lookahead picks and, in brackets, the truth it loses against the truth's best (pp); bold = the truth's pick.

LR, interval	truth best	linear to 0, main LR only	linear to 0, all LRs	searched m, main LR only	searched m main + tied m others	searched m for each LR
conv, 1	0.13	0.078 (0.02)	0.078 (0.02)	**0.13 (0.00)**	**0.13 (0.00)**	**0.13 (0.00)**
conv, 2	0.078	0.047 (0.11)	0.047 (0.11)	0.047 (0.11)	0.047 (0.11)	0.047 (0.11)
conv, 3	0.047	0.078 (0.12)	0.078 (0.12)	0.078 (0.12)	**0.047 (0.00)**	**0.047 (0.00)**
conv, 4	0.078	**0.078 (0.00)**	**0.078 (0.00)**	0.13 (0.14)	0.13 (0.14)	0.13 (0.14)
head, 1	770	460 (0.24)	460 (0.24)	1300 (0.12)	1300 (0.12)	1300 (0.12)
head, 2	460	1300 (0.18)	1300 (0.18)	1300 (0.18)	170 (0.12)	770 (0.01)
head, 3	2100	460 (0.10)	**2100 (0.00)**	460 (0.10)	770 (0.10)	770 (0.10)
head, 4	2100	1300 (0.11)	770 (0.07)	460 (0.03)	1300 (0.11)	**2100 (0.00)**
norm bias, 1	99	60 (0.51)	**99 (0.00)**	170 (0.38)	170 (0.38)	170 (0.38)
norm bias, 2	60	21 (0.06)	36 (0.03)	99 (0.11)	13 (0.12)	13 (0.12)
norm bias, 3	99	170 (0.13)	170 (0.13)	170 (0.13)	170 (0.13)	170 (0.13)
norm bias, 4	99	21 (0.01)	21 (0.01)	21 (0.01)	**99 (0.00)**	170 (0.03)
whitening bias, 1	36	21 (0.09)	**36 (0.00)**	21 (0.09)	60 (0.33)	60 (0.33)
whitening bias, 2	7.7	1.7 (0.46)	**7.7 (0.00)**	**7.7 (0.00)**	**7.7 (0.00)**	**7.7 (0.00)**
whitening bias, 3	13	21 (0.20)	60 (0.04)	36 (0.14)	60 (0.04)	60 (0.04)
whitening bias, 4	21	**21 (0.00)**	**21 (0.00)**	**21 (0.00)**	36 (0.02)	36 (0.02)

Table 3.4m. Searched cooldown multipliers over all candidates of intervals 1-4: how often 0.22 is chosen, and the most common other values.

Main LR	searched m, main only	main + tied: main m	main + tied: others' m	each: main m
conv	0.22 in 4/20; else 0.36: 7, 0.13: 4, 0.078: 2	0.22 in 6/20; else 0.36: 7, 0.13: 3, 0.078: 2	0.22 in 4/20; else 1: 3, 0.6: 3, 0.36: 3	0.22 in 6/20; else 0.36: 7, 0.13: 3, 0.078: 2
head	0.22 in 7/20; else 0.078: 5, 0.36: 3, 0.6: 3	0.22 in 7/20; else 0.36: 6, 1.7: 2, 0.6: 2	0.22 in 8/20; else 0.36: 6, 0.13: 5, 0.6: 1	0.22 in 4/20; else 0.6: 6, 0.36: 5, 1.7: 2
norm bias	0.22 in 2/20; else 0.36: 4, 0.13: 3, 0.6: 3, 7.7: 2	0.22 in 3/20; else 0.6: 5, 1: 3, 1.7: 2	0.22 in 4/20; else 0.36: 8, 0.13: 6, 0.6: 2	0.22 in 5/20; else 0.6: 6, 1.7: 2, 0.078: 2
whitening bias	0.22 in 6/20; else 0.6: 5, 0.078: 2, 0.13: 2	0.22 in 3/20; else 0.6: 5, 0.36: 3, 0.078: 2	0.22 in 4/20; else 0.36: 8, 0.13: 5, 0.6: 2	0.22 in 3/20; else 0.6: 6, 0.13: 5, 0.36: 4

3.5 inverted wave

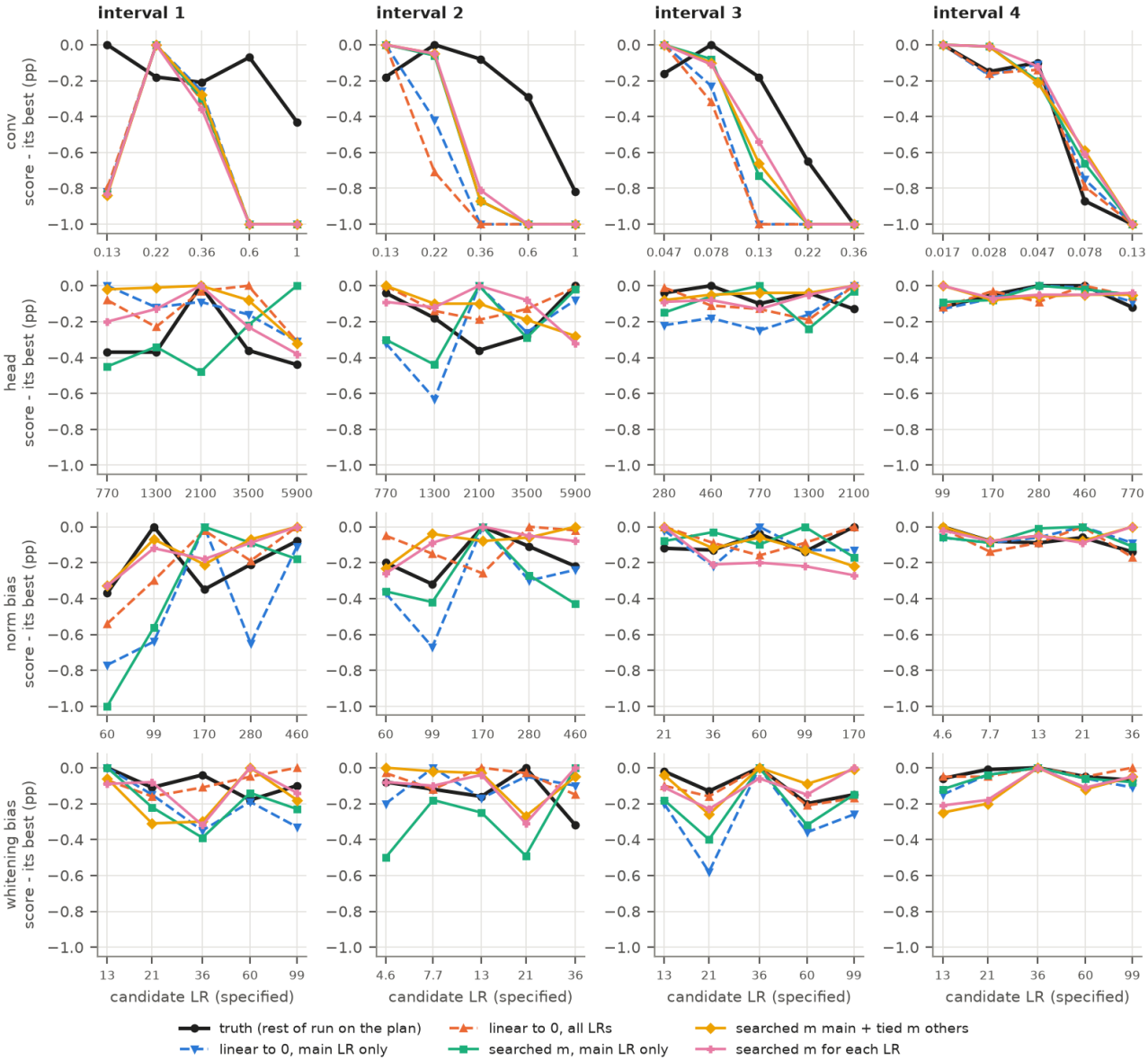


Figure 3.5. inverted wave: each score per candidate minus its own best (clipped at -1 pp); rows are the main LR, columns the interval.

Table 3.5. Candidate each lookahead picks and, in brackets, the truth it loses against the truth's best (pp); bold = the truth's pick.

LR, interval	truth best	linear to 0, main LR only	linear to 0, all LRs	searched m, main LR only	searched m main + tied m others	searched m for each LR
conv, 1	0.13	0.22 (0.18)	0.22 (0.18)	0.22 (0.18)	0.22 (0.18)	0.22 (0.18)
conv, 2	0.22	0.13 (0.18)	0.13 (0.18)	0.13 (0.18)	0.13 (0.18)	0.13 (0.18)
conv, 3	0.078	0.047 (0.16)	0.047 (0.16)	0.047 (0.16)	0.047 (0.16)	0.047 (0.16)
conv, 4	0.017	**0.017 (0.00)**	**0.017 (0.00)**	**0.017 (0.00)**	**0.017 (0.00)**	**0.017 (0.00)**
head, 1	2100	770 (0.37)	3500 (0.36)	5900 (0.44)	**2100 (0.00)**	**2100 (0.00)**
head, 2	5900	2100 (0.36)	770 (0.04)	2100 (0.36)	770 (0.04)	2100 (0.36)
head, 3	460	2100 (0.13)	2100 (0.13)	770 (0.10)	2100 (0.13)	2100 (0.13)
head, 4	280	**280 (0.00)**	460 (0.00)	**280 (0.00)**	99 (0.12)	99 (0.12)
norm bias, 1	99	170 (0.35)	460 (0.08)	170 (0.35)	460 (0.08)	460 (0.08)
norm bias, 2	170	**170 (0.00)**	280 (0.11)	**170 (0.00)**	460 (0.22)	**170 (0.00)**
norm bias, 3	170	60 (0.04)	21 (0.12)	99 (0.14)	21 (0.12)	21 (0.12)
norm bias, 4	4.6	21 (0.06)	21 (0.06)	21 (0.06)	**4.6 (0.00)**	36 (0.14)
whitening bias, 1	13	**13 (0.00)**	99 (0.10)	**13 (0.00)**	60 (0.18)	60 (0.18)
whitening bias, 2	21	7.7 (0.12)	13 (0.16)	36 (0.32)	4.6 (0.08)	36 (0.32)
whitening bias, 3	36	**36 (0.00)**	**36 (0.00)**	**36 (0.00)**	**36 (0.00)**	99 (0.15)
whitening bias, 4	36	**36 (0.00)**	**36 (0.00)**	**36 (0.00)**	**36 (0.00)**	**36 (0.00)**

Table 3.5m. Searched cooldown multipliers over all candidates of intervals 1-4: how often 0.22 is chosen, and the most common other values.

Main LR	searched m, main only	main + tied: main m	main + tied: others' m	each: main m
conv	0.22 in 4/20; else 0.13: 5, 0.078: 5, 0.36: 3	0.22 in 3/20; else 0.078: 6, 0.13: 5, 0.36: 4	0.22 in 3/20; else 1.7: 4, 1: 3, 0.6: 3, 0.36: 3	0.22 in 3/20; else 0.078: 6, 0.13: 5, 0.36: 4
head	0.22 in 4/20; else 0.6: 5, 0.078: 5, 0.13: 2	0.22 in 7/20; else 4.6: 3, 0.6: 3, 0.13: 2	0.22 in 8/20; else 0.078: 7, 0.13: 2, 0.36: 2	0.22 in 3/20; else 0.36: 4, 1: 4, 0.13: 3
norm bias	0.22 in 5/20; else 0.36: 4, 0.13: 3, 0.6: 3	0.22 in 5/20; else 0.13: 4, 0.36: 3, 0.078: 2	0.22 in 8/20; else 0.078: 6, 0.047: 2, 0.13: 2	0.22 in 2/20; else 0.36: 4, 0.078: 4, 0.13: 4, 0.6: 3
whitening bias	0.22 in 6/20; else 0.36: 4, 0.6: 2, 1: 2	0.22 in 7/20; else 0.36: 4, 0.028: 3, 0.078: 3	0.22 in 7/20; else 0.078: 10, 0.36: 3	0.22 in 7/20; else 0.078: 4, 0.047: 2, 0.36: 2

3.6 steep ramp x1/8->x8

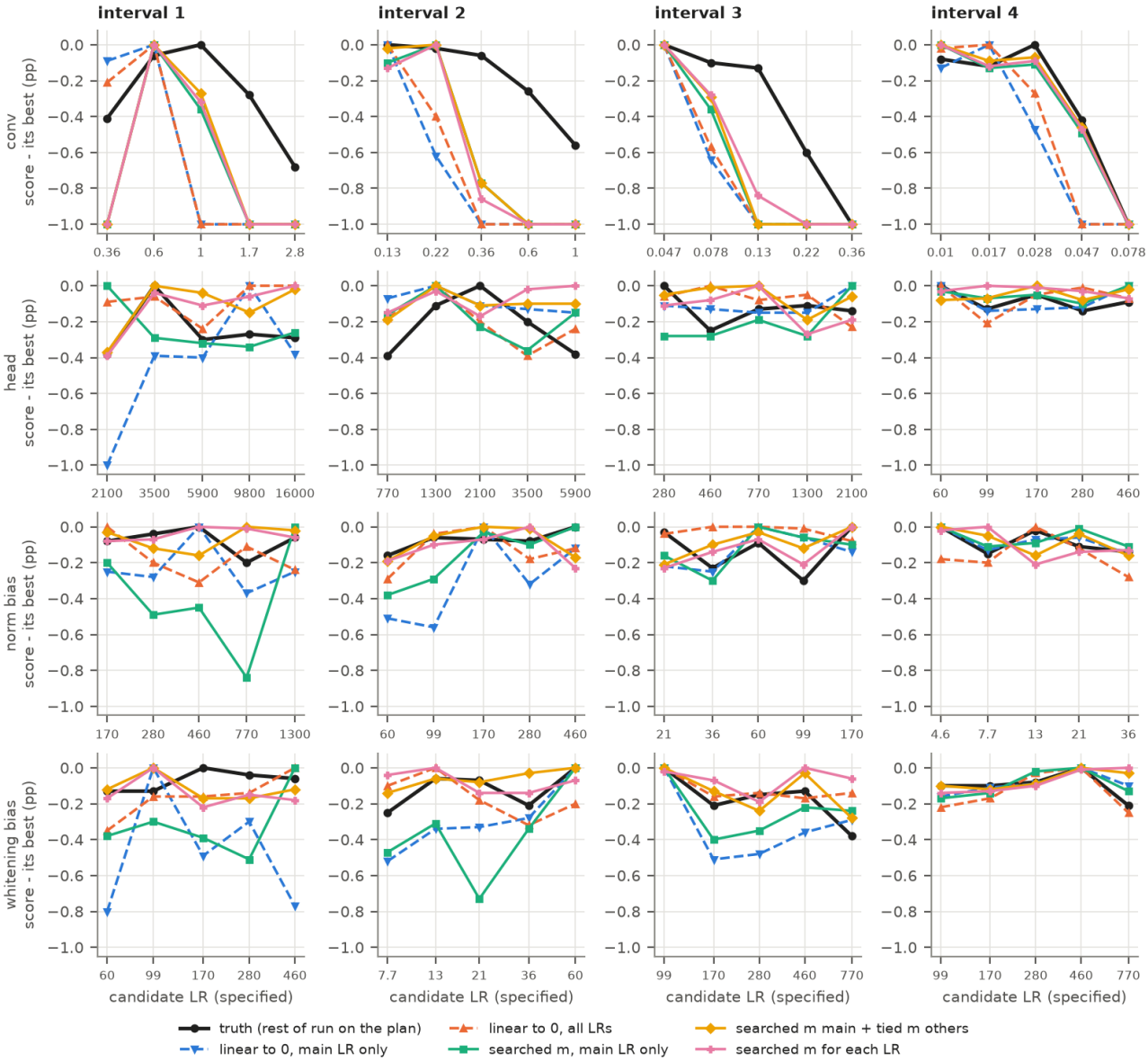


Figure 3.6. steep ramp x1/8->x8: each score per candidate minus its own best (clipped at -1 pp); rows are the main LR, columns the interval.

Table 3.6. Candidate each lookahead picks and, in brackets, the truth it loses against the truth's best (pp); bold = the truth's pick.

LR, interval	truth best	linear to 0, main LR only	linear to 0, all LRs	searched m, main LR only	searched m main + tied m others	searched m for each LR
conv, 1	1	0.6 (0.06)	0.6 (0.06)	0.6 (0.06)	0.6 (0.06)	0.6 (0.06)
conv, 2	0.13	**0.13 (0.00)**	**0.13 (0.00)**	0.22 (0.02)	0.22 (0.02)	0.22 (0.02)
conv, 3	0.047	**0.047 (0.00)**	**0.047 (0.00)**	**0.047 (0.00)**	**0.047 (0.00)**	**0.047 (0.00)**
conv, 4	0.028	0.017 (0.12)	0.017 (0.12)	0.01 (0.08)	0.01 (0.08)	0.01 (0.08)
head, 1	3500	9800 (0.27)	9800 (0.27)	2100 (0.38)	**3500 (0.00)**	16000 (0.29)
head, 2	2100	1300 (0.11)	1300 (0.11)	1300 (0.11)	1300 (0.11)	5900 (0.38)
head, 3	280	2100 (0.14)	460 (0.25)	2100 (0.14)	770 (0.13)	770 (0.13)
head, 4	60	**60 (0.00)**	**60 (0.00)**	460 (0.09)	170 (0.05)	99 (0.13)
norm bias, 1	460	**460 (0.00)**	170 (0.08)	1300 (0.06)	770 (0.20)	**460 (0.00)**
norm bias, 2	460	170 (0.07)	170 (0.07)	**460 (0.00)**	170 (0.07)	280 (0.08)
norm bias, 3	170	60 (0.09)	36 (0.23)	60 (0.09)	**170 (0.00)**	**170 (0.00)**
norm bias, 4	4.6	**4.6 (0.00)**	13 (0.02)	**4.6 (0.00)**	**4.6 (0.00)**	7.7 (0.15)
whitening bias, 1	170	99 (0.13)	460 (0.06)	460 (0.06)	99 (0.13)	99 (0.13)
whitening bias, 2	60	**60 (0.00)**	13 (0.06)	**60 (0.00)**	**60 (0.00)**	13 (0.06)
whitening bias, 3	99	**99 (0.00)**	**99 (0.00)**	**99 (0.00)**	**99 (0.00)**	460 (0.13)
whitening bias, 4	460	**460 (0.00)**	**460 (0.00)**	**460 (0.00)**	**460 (0.00)**	770 (0.21)

Table 3.6m. Searched cooldown multipliers over all candidates of intervals 1-4: how often 0.22 is chosen, and the most common other values.

Main LR	searched m, main only	main + tied: main m	main + tied: others' m	each: main m
conv	0.22 in 0/20; else 0.078: 11, 0.13: 5, 0.047: 3, 0.028: 1	0.22 in 0/20; else 0.078: 12, 0.13: 5, 0.047: 3	0.22 in 6/20; else 0.36: 4, 0.078: 3, 0.13: 3	0.22 in 0/20; else 0.078: 13, 0.13: 4, 0.047: 3
head	0.22 in 4/20; else 0.6: 6, 0.13: 4, 0.078: 2	0.22 in 1/20; else 0.13: 3, 0.047: 3, 0.36: 3, 0.078: 3	0.22 in 0/20; else 0.047: 14, 0.078: 5, 0.028: 1	0.22 in 3/20; else 0.078: 5, 0.6: 5, 0.36: 3
norm bias	0.22 in 6/20; else 0.6: 4, 0.028: 3, 0.078: 3	0.22 in 1/20; else 0.13: 7, 0.36: 4, 0.6: 3, 1: 2	0.22 in 0/20; else 0.047: 10, 0.078: 10	0.22 in 4/20; else 0.36: 6, 0.13: 5, 0.078: 2
whitening bias	0.22 in 4/20; else 0.13: 7, 0.078: 4, 0.36: 3	0.22 in 2/20; else 0.13: 6, 0.36: 5, 0.6: 2	0.22 in 0/20; else 0.047: 12, 0.078: 8	0.22 in 4/20; else 0.36: 4, 1.7: 3, 0.6: 3